

Konstantin Dimitriadis Lorenz

Email: boid@tutamail.com Phone number: +46 72-168 40 58 GitHub: kodi1046

LinkedIn: Konstantin Dimitriadis Lorenz Location: Uppsala, Sweden

Education

University

Uppsala University, B.S. Computer Science, CGPA 3.97

2024 – 2027 expected

Relevant Courses (UU)

- **Math** Linear Algebra, Multivariable Calculus, Probability Theory 2026
- **Computer Science** OOP, Algorithmic Theory, (C, Java, Python) 2025

Experience

StuddyBuddy

summer 2025 – winter 2026

A tutoring business for High School students. Taught mathematics to students across 3 grade levels, and improved their academic performance through tailored lesson plans.

Rookie Startups

summer 2024

A summer job in which I built a React-based web application to estimate CO₂ emissions of food products. Implemented a dynamic search and data visualization.

Projects

HVol

winter 2026 – present

Developing a Heston-Stochastic-Volatility simulator in C++ for option pricing. Includes the Carr-Madan quasi-closed-form solution for European options, Monte-Carlo simulation for Vanilla, and Exotic options. FFT implementation(s) (radix-2 DIT), and Heston parameter calibration.

Quant.py

winter 2026 – present

Developing Quant.py, a modular Python backtesting engine for trading strategies and option portfolios. Implemented pluggable pricing engines (Black-Scholes, Monte Carlo), GBM, Jump-diffusion simulation, delta hedging, Greeks calculation, and portfolio tracking over historical or synthetic data.

Reference-counting Garbage Collector

winter 2025 – 2026

A joint project in which we implemented a reference-counting garbage collector in C.

Symbolic Calculator

winter 2025

A two person project where we built a symbolic calculator in Java using Pratt parsing, utilizing the visitor pattern for AST traversals.

Minesweeper

spring 2025

A joint project where we created Minesweeper, as a terminal-based game in TypeScript.

Chat App

fall 2024

A two person project, wherein we created a chat application in Django (python) with websockets.

Space Game

spring 2024

A space-themed 2D single-player story-game created in Godot.

Research

The Uncertainty Principle in Time-Series Momentum

May 2026

A paper applying the Gabor–Heisenberg uncertainty principle to time-series momentum strategies. Derives irreducible lower bounds on signal lag, portfolio turnover, and regime-detection delay as direct consequences of the time-frequency resolution limit of linear filters.

Skills

'*' denotes academic or self-study experience

- **Programming**

- **python:** pytorch*, Numpy, Matplotlib, Pandas, Django
- **JavaScript/TypeScript:** Jest, React, Svelte*
- **C:** CUnit, GDB
- **Java:** JUnit
- **C++***
- **Misc:** Make, Linux, Git, Github

- **Languages (CEFR)**

- **Swedish:** C2
- **English:** C1/C2, IELTS Academic overall band 8 (2025)
- **German:** C1
- **Spanish:** C1
- **Japanese:** B2